

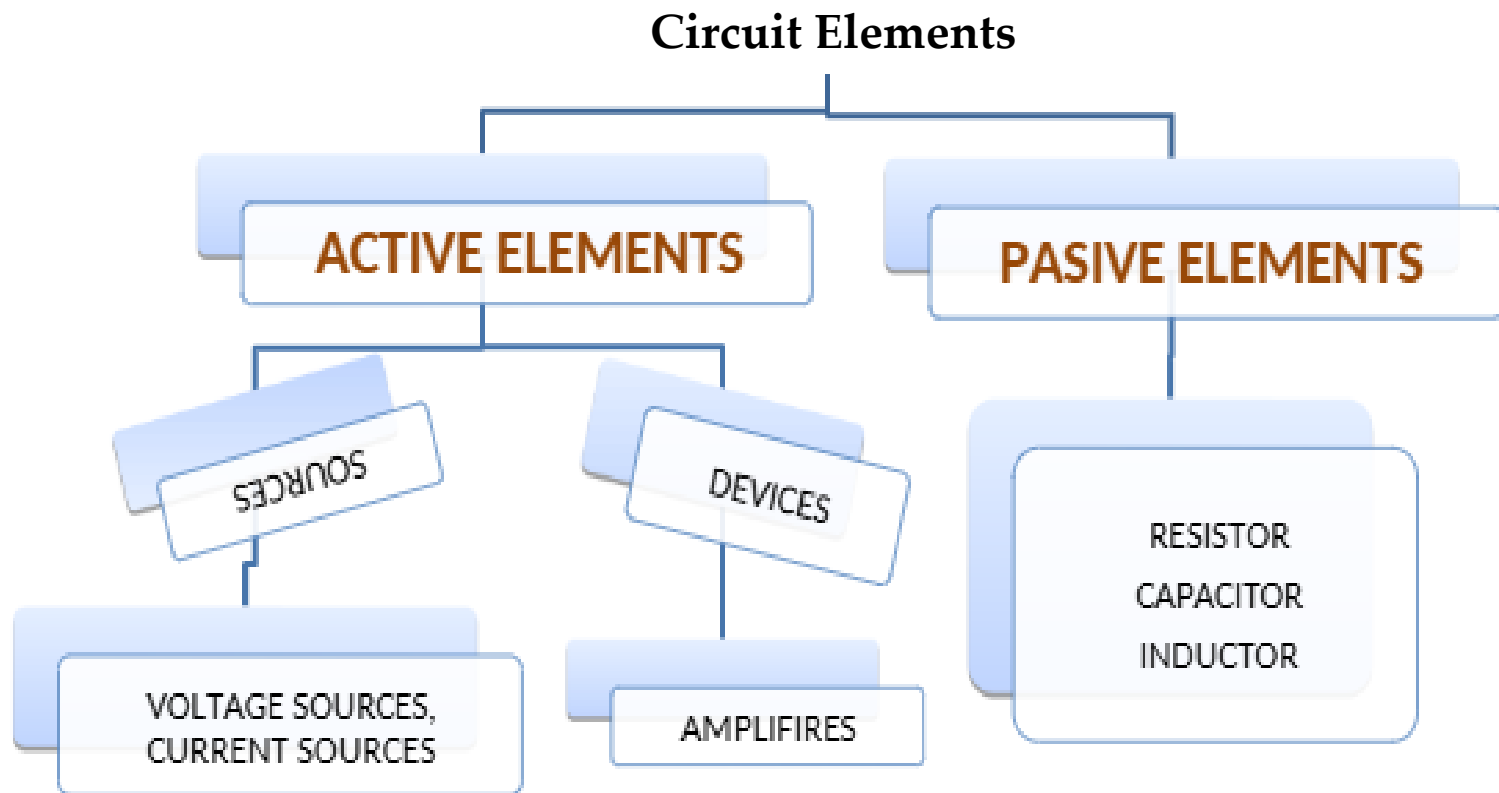
Module 1

DC Circuit Analysis and Theorems

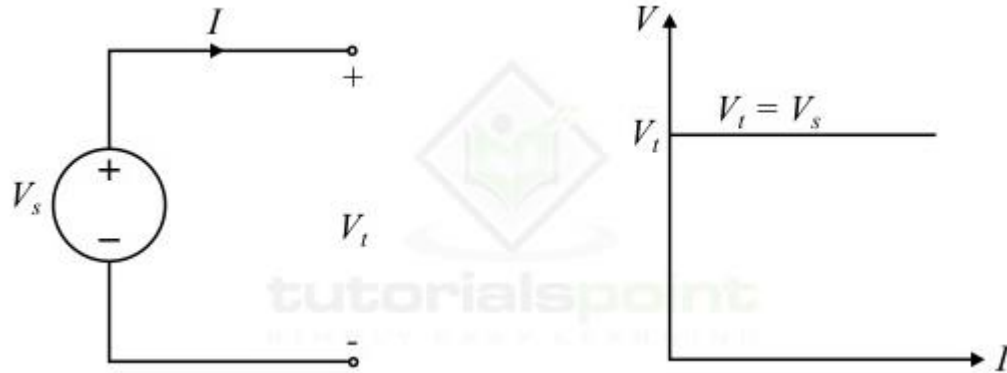
Dr. Elizabeth Rufus

Professor, School of Electronics Engineering

Circuit Elements

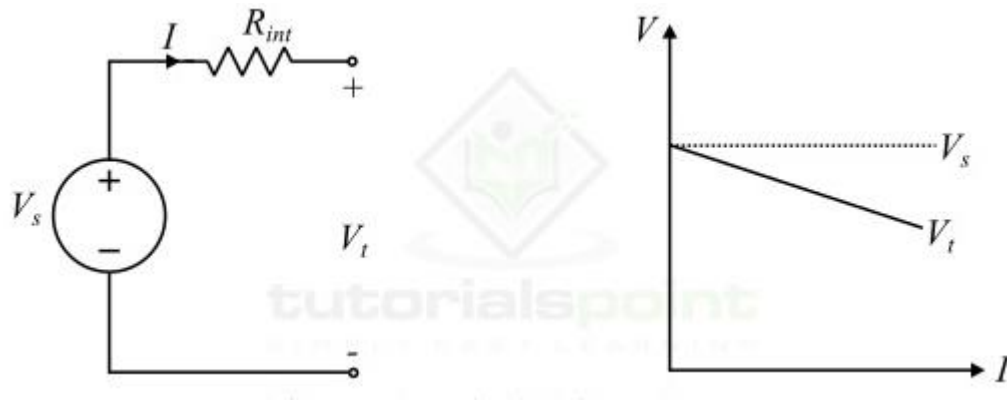


Voltage Source



Ideal Voltage Source

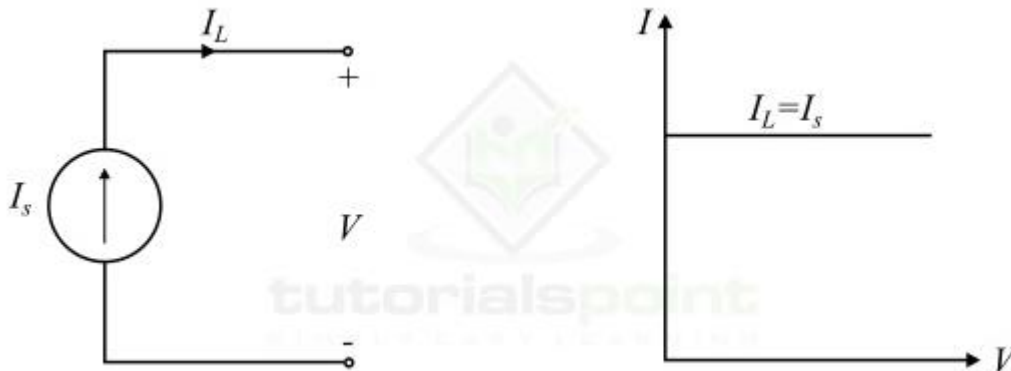
A voltage source which has zero internal resistance is known as ideal voltage source.



Practical Voltage Source

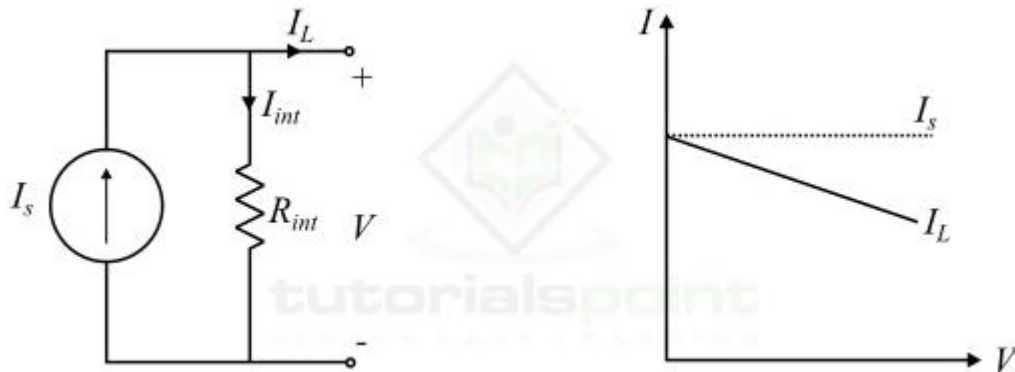
An independent voltage source which has a finite internal resistance is known as a **practical voltage source** or **real voltage source**. When a current is drawn from the source, it causes a voltage drop in the internal resistance. Consequently, the output voltage at the source terminals is reduced.

Current Source



Ideal Current Source

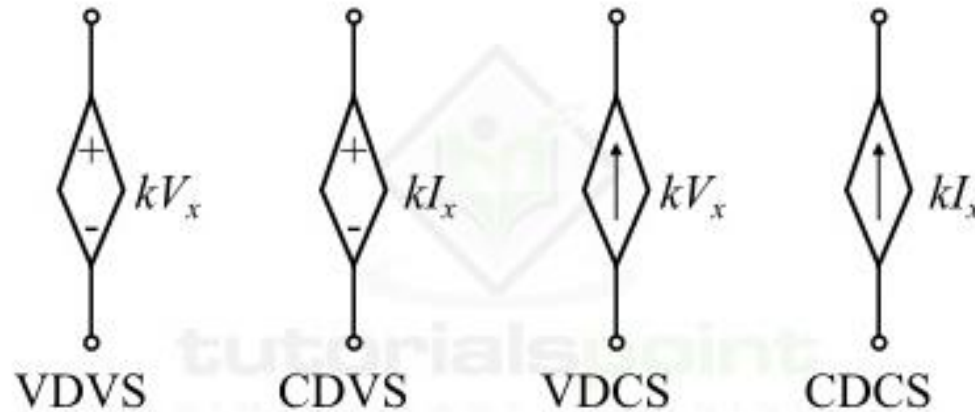
A current source which has **infinite internal resistance** is known as **ideal current source**.



Practical Current Source

A **practical current source (or real current source)** is one which has a finite internal resistance which is connected in parallel with the source current. Therefore, when there is any change in the load voltage, the current delivered by the real current source is changed.

Dependent sources



Voltage Dependent Voltage Source (VDVS)

Current Dependent Voltage Source (CDVS)

Voltage Dependent Current Source (VDCS)

Current Dependent Current Source (CDCS)

Resistor

Offers resistance in any electrical circuit or network



American-style symbols. (a) resistor,
(b) rheostat (variable resistor), and
(c) potentiometer



IEC-style resistor symbol

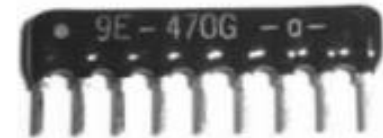
[*The **International Electrotechnical Commission**(IEC; *Commission électrotechnique internationale* (CEI), in French) is a non-profit, non-governmental international standards organization that prepares and publishes International Standards for all electrical, electronic and related technologies - collectively known as "electrotechnology"]



Showing a resistor(axial lead resistor) component with $330\ \Omega$ and a tolerance of 5%



An aluminium-housed power resistor rated for 50 W when heat-sinked



A single in line (SIL) resistor package with 8 individual, 47 ohm resistors. One end of each resistor is connected to a separate pin and the other ends are all connected together to the remaining (common) pin - pin 1, at the end identified by the white dot.



Resistors with wire leads for through-hole mounting



Three carbon composition resistors in a 1960s valve (vacuum tube) radio



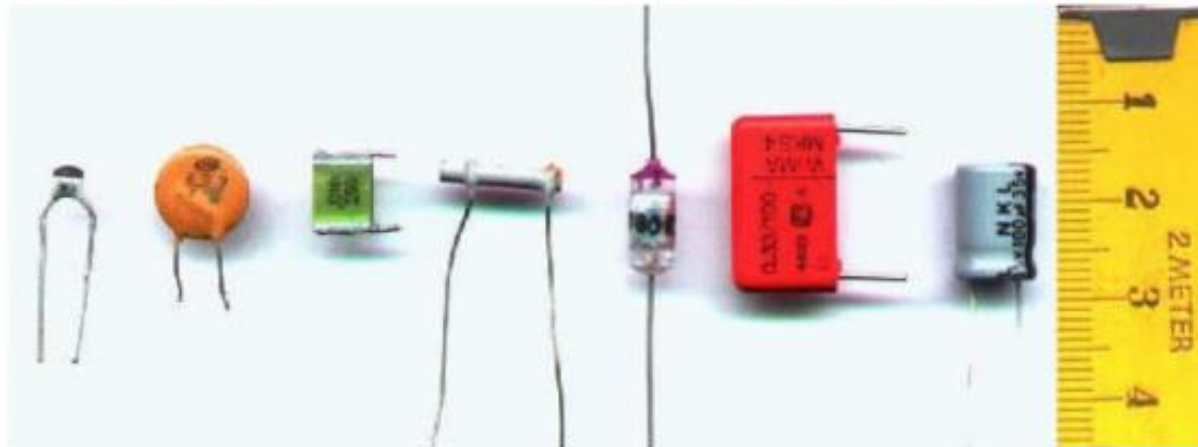
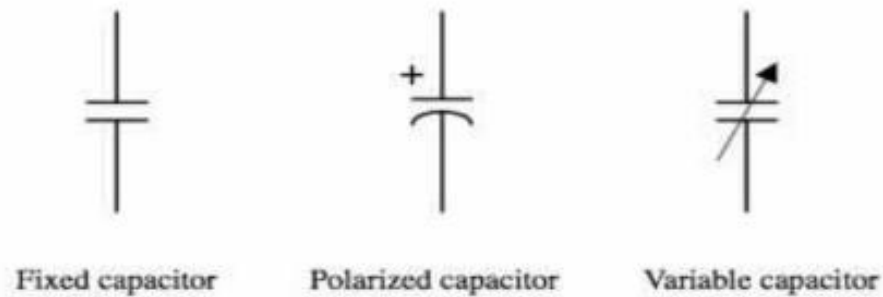
Partially exposed Tesla TR-212 $1\ \text{k}\Omega$ carbon film resistor



A carbon resistor printed directly onto the SMD pads on a PCB

Capacitor

A passive two-terminal electrical component used to store energy electrostatically in an electric field. It is characterized by capacitance.



Different types of capacitors: From left: multilayer ceramic, ceramic disc, multilayer polyester film, tubular ceramic, polystyrene, metalized polyester film, aluminum electrolytic. Major scale divisions are in centimeters.

Inductor

A passive two-terminal electrical component which resists changes in electric current passing through it. When a current flows through it, energy is stored temporarily in a magnetic field in the coil.



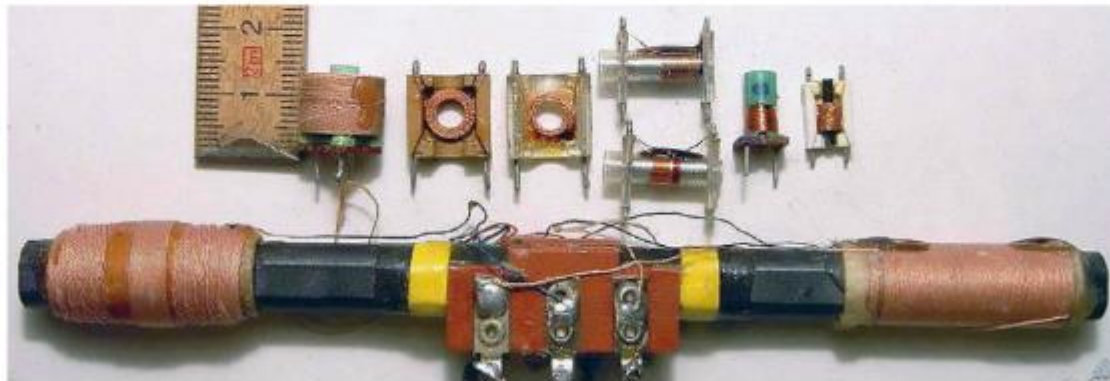
A selection of low-value inductors



Axial lead inductors (100 μ H)



A ferrite "bead" choke, consisting of an encircling ferrite cylinder, removes electronic noise from a computer power cord.



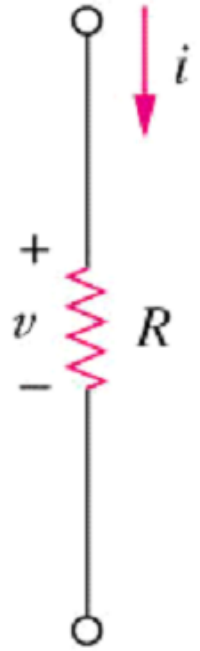
Ferrite rod antenna [loopstick](#) (bottom) and RF chokes, coils and IF transformers (top) used in radio equipment. The ferrite antenna has two windings, to receive both the LW and MW bands. The loopstick and chokes at top right are wound with [litz wire](#) in a [basket weave](#) pattern, with adjacent winding layers at different angles, to reduce resistance due to [proximity effect](#) and [skin effect](#).

Ohms Law

- Ohm's law states that the voltage across a resistor is directly proportional to the current I flowing through the resistor.
- Mathematical expression for Ohm's Law is as follows:

$$v = iR$$

- Two extreme possible values of $R = 0$ (**zero**) and ∞ (**infinite**) are related with two basic circuit concepts: **short circuit** and **open circuit**.



- Conductance is the ability of an element to conduct electric current; it is the reciprocal of resistance R and is measured in mhos or Siemens.

$$G = \frac{1}{R} = \frac{i}{v}$$

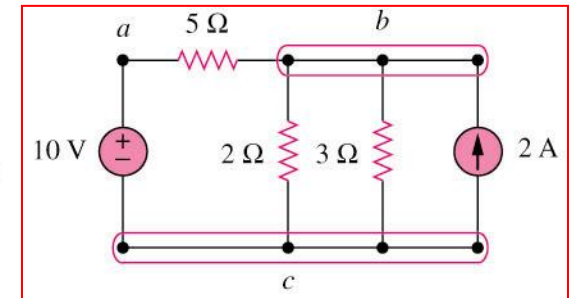
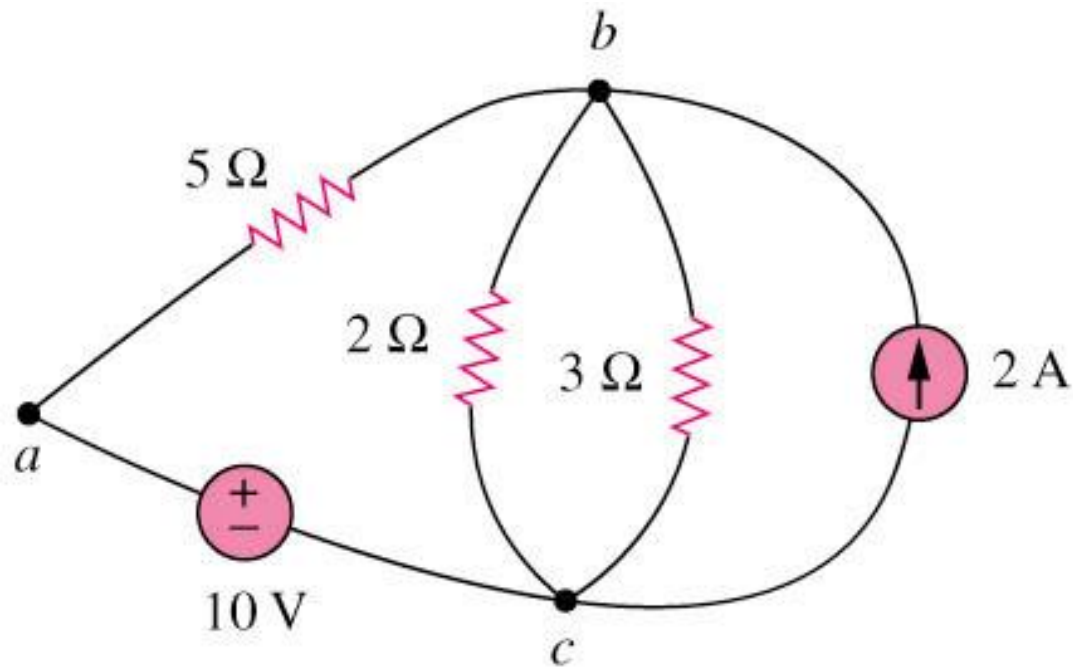
- The power dissipated by a resistor:

$$p = vi = i^2 R = \frac{v^2}{R}$$

Nodes, Branches and Loops

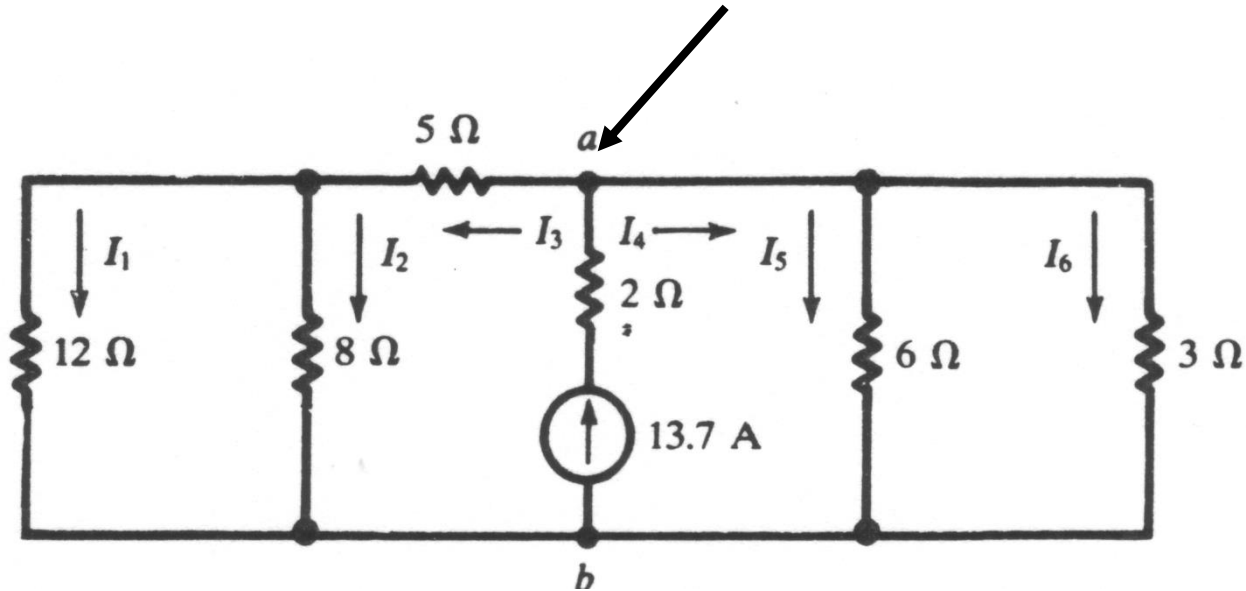
- A **branch** represents a single element such as a voltage source or a resistor.
- A **node** is the point of connection between two or more branches.
- A **loop** is any closed path in a circuit.
- A network with b branches, n nodes, and l independent loops will satisfy the fundamental theorem of network topology:

$$b = l + n - 1$$



How many branches, nodes and loops are there?

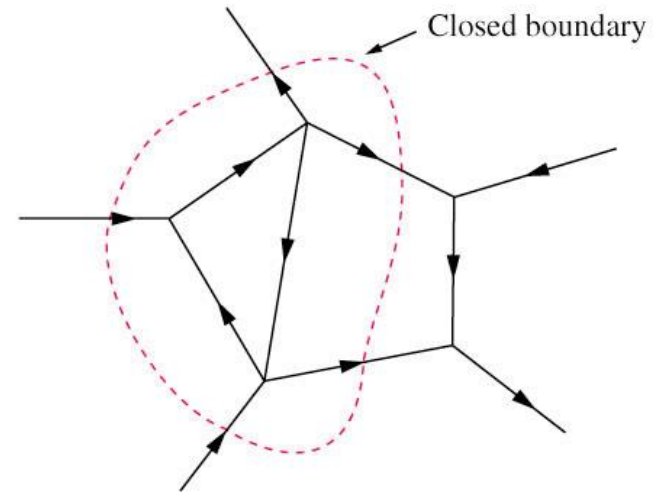
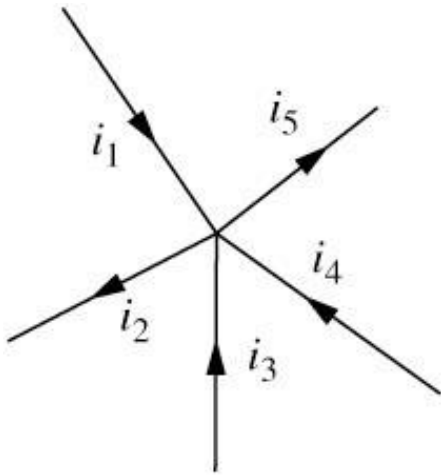
Should we consider it as one branch or two branches?



How many branches, nodes and loops are there?

Kirchhoff's Current Law

- Kirchhoff's current law (KCL) states that the algebraic sum of currents entering a node (or a closed boundary) is zero.

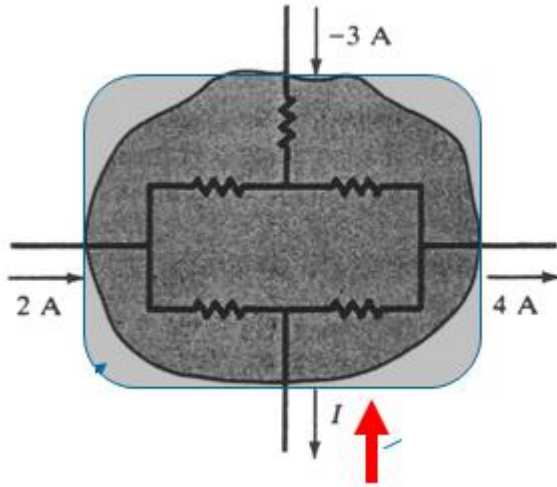


Mathematically,

$$\sum_{n=1}^N i_n = 0$$

Kirchhoff's Current Law

Determine the current I for the circuit shown in the figure below.



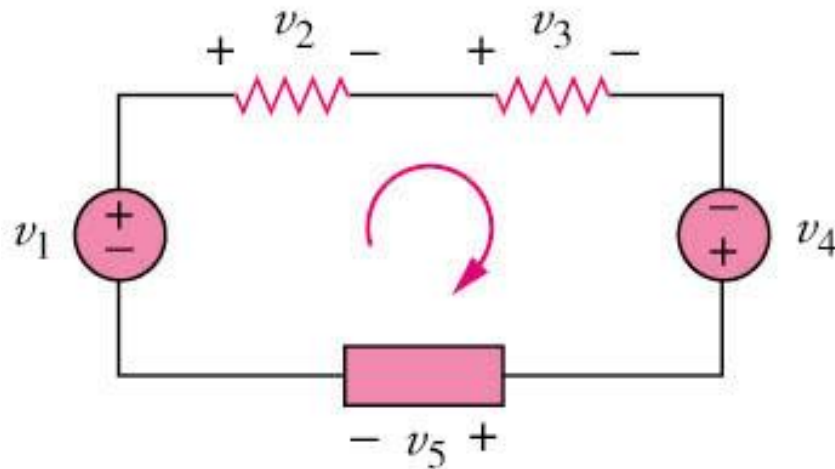
$$I + 4 - (-3) - 2 = 0$$

$$\Rightarrow I = -5\text{ A}$$

This indicates that the actual current for I is flowing in the opposite direction.

Kirchhoff's Voltage Law

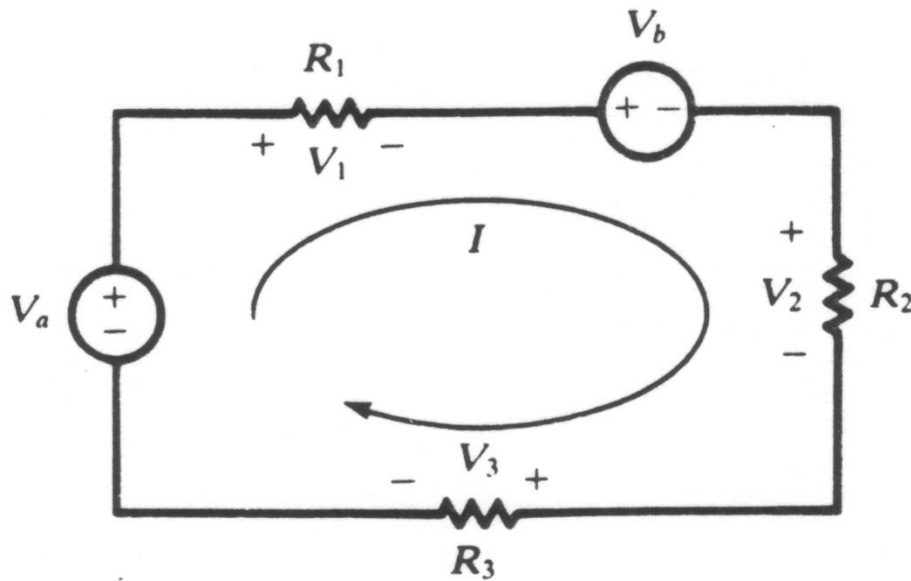
Kirchhoff's voltage law (KVL) states that the algebraic sum of all voltages around a closed path (or loop) is zero.



Mathematically,
$$\sum_{m=1}^M v_n = 0$$

Kirchhoff's Voltage Law

Applying the KVL equation for the circuit in the figure below.



$$V_a - V_1 - V_b - V_2 - V_3 = 0$$

$$V_1 = IR_1 \quad V_2 = IR_2 \quad V_3 = IR_3$$

$$\Rightarrow V_a - V_b = I(R_1 + R_2 + R_3)$$

$$I = \frac{V_a - V_b}{R_1 + R_2 + R_3}$$

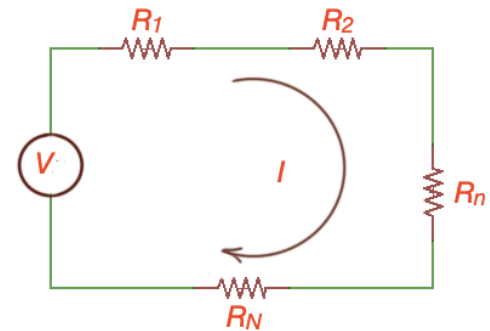
Series Resistors and Voltage Division

- Series: Two or more elements are in series if they are cascaded or connected sequentially and consequently carry the same current.
- The equivalent resistance of any number of resistors connected in a series is the sum of the individual resistances.

$$R_{eq} = R_1 + R_2 + \cdots + R_N = \sum_{n=1}^N R_n$$

- The voltage divider can be expressed as

$$V_n = \frac{R_n}{R_1 + R_2 + \cdots + R_N} V$$



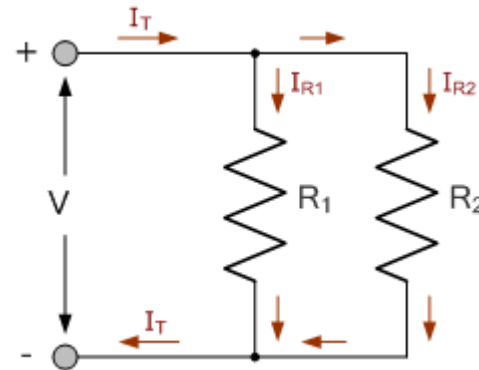
Parallel Resistors and Current Division

- Parallel: Two or more elements are in parallel if they are connected to the same two nodes and consequently have the same voltage across them.
- The equivalent resistance of a circuit with N resistors in parallel is:

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_N}$$

- The total current i is shared by the resistors in inverse proportion to their resistances. The current divider can be expressed as:

$$i_n = \frac{v}{R_n} = \frac{i R_{eq}}{R_n}$$



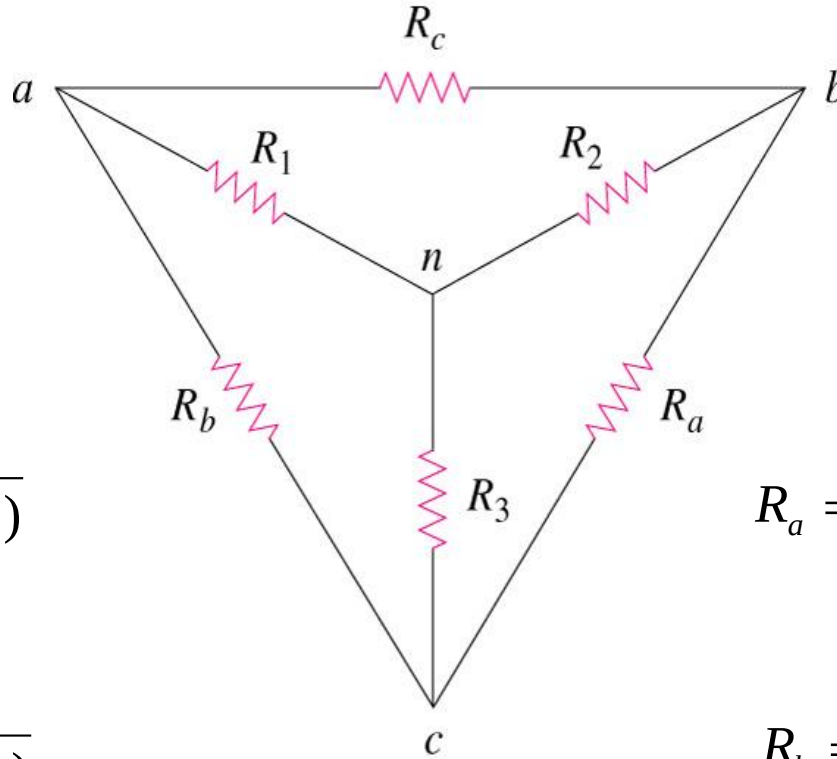
Wye-Delta Transformations

Delta -> Star

$$R_1 = \frac{R_b R_c}{(R_a + R_b + R_c)}$$

$$R_2 = \frac{R_c R_a}{(R_a + R_b + R_c)}$$

$$R_3 = \frac{R_a R_b}{(R_a + R_b + R_c)}$$



Star -> Delta

$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1}$$

$$R_b = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2}$$

$$R_c = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3}$$

Next class: Nodal Analysis